AP CALCULUS AB

Unit 1 • Worksheet 1-3A

Worksheet 1-3A • Limits to Infinity and Horizontal Asymptotes

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which describes the graph of $f(x) = \dfrac{x+1}{x^2-4x-5}$?

(A) A hole at $x = -1$ and a vertical asymptote at $x = 5$

(B) Vertical asymptotes at $x = -1$ and $x = 5$

(C) A hole at $x = 5$ and a vertical asymptote at $x = -1$

(D) No hole and no vertical asymptote

Question 2

NO CALCULATOR

What is the y-coordinate of the hole in the graph of $f(x) = \dfrac{x+1}{x^2-4x-5}$?

(A) 0

(B) $-\frac{1}{6}$

(C) $\frac{1}{6}$

(D) $-\frac{1}{4}$

Question 3

NO CALCULATOR

For the same f, what is $\lim_{x \to 5^+} f(x)$?

(A) $-\infty$

(B) 0

(C) $+\infty$

(D) $\frac{1}{6}$

Question 4

NO CALCULATOR

What is $\lim_{x\to\infty} \dfrac{2x^2-3}{5x^2+x}$?

(A) $\frac{2}{5}$
(B) 0
(C) -3
(D) It does not exist.

Question 5

NO CALCULATOR

What is $\lim_{x\to\infty} \dfrac{8x-1}{2x^3+5}$?

(A) 4
(B) 8
(C) It does not exist.
(D) 0

Question 6

NO CALCULATOR

What is $\lim_{x\to-\infty} \dfrac{x^3+1}{4x-x^2}$?

(A) -1
(B) It does not exist (the function grows without bound).
(C) 0
(D) $\frac{1}{4}$

Question 7

NO CALCULATOR

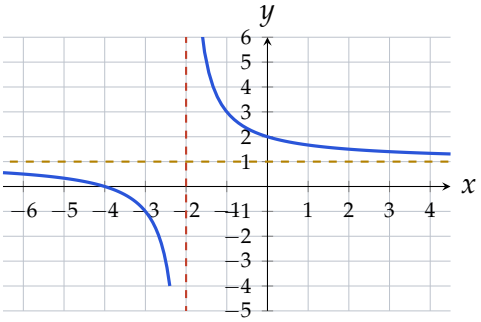
Which are all the asymptotes of $y = \dfrac{3x^2+1}{x^2-9}$?

(A) $x = 3$ and $y = 3$ only
(B) $x = -3$, $x = 3$ and $y = 0$
(C) $x = -3$, $x = 3$ and $y = 3$
(D) $y = 3$ only

Question 8

NO CALCULATOR

The graph of g is shown, with its asymptotes. What is $\lim_{x\to-2^-} g(x)$?



(A) 1
(B) $+\infty$
(C) 0
(D) $-\infty$

Question 9 (Free response) NO CALCULATOR

Let $k(x) = \dfrac{2x^2 - 2}{x^2 + 2x - 3}$.

(a) Factor k. Find the vertical asymptote and the hole (both coordinates).
(b) Find $\lim_{x\to\infty} k(x)$ and $\lim_{x\to-\infty} k(x)$. What is the horizontal asymptote?
(c) Find $\lim_{x\to-3^+} k(x)$.